

SUPPLEMENTARY MATERIAL

Critical Fragility Emerges from Chromosomal Instability in Cancer

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1 Stationary Mutational Population distributions in the Liquid Tumor case

We consider a liquid tumor model with fixed population size N . The number of wild-type cells is indicated with W , the number of dead cells with D . The cancer cell population can be further divided in N_I sub-populations (one per instability class), with N_I the number of instability genes. We indicate the abundance of each cancer sub-population with C_k , where $k \in \{1, \dots, N_I\}$ counts the number of mutated instability genes. We work with densities (fractions) rather than absolute abundances:

$$f_W \equiv \frac{W}{N}, \quad f_k \equiv \frac{C_k}{N}, \quad p_D \equiv \frac{D}{N} \quad (1)$$

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These are constrained by the fixed-size normalization:

$$\sum_{k=1}^{N_I} f_k + f_W + p_D = 1 \quad (2)$$

Survival probability after a mutation round

Both cancer and wild-type cells undergo replication followed by a stochastic mutation event. After mutation, a viability check is performed on all N_{HK} housekeeping genes: if any housekeeping gene becomes fully mutated (on all chromosome copies), the cell dies. Each gene has an independent per-locus mutation probability $\mu_k = k \cdot \delta\mu$ for a class- k cell. The probability that a single housekeeping gene survives (i.e. is not mutated) in one round is $(1 - k \delta\mu)$. Since all N_{HK} housekeeping genes must independently survive:

$$P_{s,k} = \underbrace{(1 - k \delta\mu)}_{\text{one HK gene survives}} \times \cdots \times \underbrace{(1 - k \delta\mu)}_{N_{HK} \text{ times}} = (1 - k \delta\mu)^{N_{HK}} \quad (3)$$

The post-replication fates (exact multi-step model)

After a mutation round, a cell in class k can either remain in class k (cloning), promote to class $k + j$ with $j \geq 1$ (multi-step promotion), or die due to a housekeeping mutation. Since mutations at different loci occur independently with per-locus probability $\mu_k = k \cdot \delta\mu$, we derive these probabilities exactly:

1. Faithful cloning (α_k): The cell remains in class k . This requires that none of the N_{HK} housekeeping genes and none of the $(N_I - k)$ remaining unmutated instability genes are hit. The total number of loci that must remain unmutated is $N_{HK} + N_I - k$:

$$\alpha_k = (1 - k \delta\mu)^{N_{HK} + N_I - k} \quad (4)$$

2. Promotion ($P_{k \rightarrow k+j}$): The cell advances from class k to class $k + j$ (where $j \in \{1, \dots, N_I - k\}$ instability genes mutate, while all other instability loci and all housekeeping loci remain unmutated). Since the selection of which instability genes mutate is combinatorial:

$$P_{k \rightarrow k+j} = \binom{N_I - k}{j} (k \delta\mu)^j (1 - k \delta\mu)^{N_{HK} + N_I - k - j} \quad (5)$$

Note that faithful cloning is simply the diagonal case $\alpha_k = P_{k \rightarrow k}$.

3. Lethal mutation (δ_k): The cell dies because at least one housekeeping gene is hit, which is the complement of the survival probability $P_{s,k} = (1 - k \delta\mu)^{N_{HK}}$:

$$\delta_k = 1 - P_{s,k} = 1 - (1 - k \delta\mu)^{N_{HK}} \quad (6)$$

The sum of all possible survival outcomes plus the death probability is exactly 1:

$$\sum_{j=0}^{N_I-k} P_{k \rightarrow k+j} + \delta_k = (1-k\delta\mu)^{N_{HK}} \sum_{j=0}^{N_I-k} \binom{N_I-k}{j} (k\delta\mu)^j (1-k\delta\mu)^{N_I-k-j} + [1 - (1-k\delta\mu)^{N_{HK}}] = 1 \quad (7)$$

This binomial formulation provides an exact description of the mutation process without needing leading-order approximations.

Moran competition probabilities

When a dividing cell targets a living (non-dead) cell, the outcome is decided by a Moran process: the divider replaces the target with probability proportional to its fitness relative to the combined fitness. All cancer classes share the same intrinsic division rate $r_{\max}$, while wild-type cells divide at rate r_0 :

- **Cancer vs Wild-Type:** The cancer cell (rate $r_{\max}$) competes against a wild-type cell (rate r_0):

$$\beta_{cw} = \frac{r_{\max}}{r_{\max} + r_0} \quad (8)$$

- **Cancer vs Cancer:** Two cancer cells with the same rate $r_{\max}$ compete symmetrically:

$$\beta_{cc} = \frac{r_{\max}}{r_{\max} + r_{\max}} = \frac{1}{2} \quad (9)$$

- **Wild-Type vs Cancer:** The complement of the cancer-vs-WT outcome:

$$\beta_{wc} = \frac{r_0}{r_{\max} + r_0} = 1 - \beta_{cw} \quad (10)$$

System parameters are formalized in Table 1.

1.1 Microscopic Mechanisms and Elementary Transition Rates

After each duplication step, both mother and daughter cells independently undergo stochastic mutation events as described above. Both cells independently draw from the same probability distribution: cloning (α_k), promoting ($P_{k \rightarrow k+j}$), or dying (δ_k). Crucially, DNA replication and subsequent viability/mutation checks are only performed upon successful division (Moran competition success). If division is aborted, neither the mother nor the daughter replicates or mutates. Thus, all division-dependent rates (cloning, promotion, and death) for both mother and daughter cells are scaled by the division success probability ϕ_k .

Table 1: Systematic Overview of the Unified Parameter Space

Symbol	Biological Definition	Functional Form / Constraint
N	Total fixed tissue lattice sites	Constant volume constraint
N_I	Maximum viable cancer mutational classes	Total mutable Instability genes
N_{HK}	Total critical Housekeeping genes	Continuous fitness check loci
$\delta\mu$	Per-locus mutation rate increment	Constant scaling factor
$r_{\max}$	Intrinsic division rate of class k cancer	Phenotypic fitness parameter
r_0	Active division rate of healthy wild-type	Homeostatic driver baseline
α_k	Probability of a faithful clone	$(1 - k \delta\mu)^{N_{HK} + N_I - k}$
$P_{k \rightarrow k+j}$	Probability of promotion ($k \rightarrow k + j$)	$\binom{N_I - k}{j} (k \delta\mu)^j (1 - k \delta\mu)^{N_{HK} + N_I - k - j}$
δ_k	Lethal mutation probability	$1 - (1 - k \delta\mu)^{N_{HK}}$
β_{cw}	Moran win probability of cancer vs WT	$r_{\max} / (r_{\max} + r_0)$
β_{cc}	Moran win probability of cancer k vs j	$r_{\max} / (r_{\max} + r_{\max})$
β_{wc}	Moran win probability of WT vs cancer k	$r_0 / (r_{\max} + r_0) = 1 - \beta_{cw}$

Effective reachable space ϕ_k

We define the fractional reachable space ϕ_k as the effective probability that a dividing class- k cancer cell successfully places its daughter. The daughter can land on three types of targets:

$$\phi_k = \underbrace{p_D}_{\text{dead slots: always available}} + \underbrace{f_W \beta_{cw}}_{\text{WT cells displaced via Moran competition}} + \underbrace{\sum_{j=1}^{N_I} f_j \beta_{cc}}_{\text{cancer cells displaced (50\% win rate)}} \quad (11)$$

- p_D : the fraction of dead (empty) slots. Replacement always succeeds (no competition needed).
- $f_W \beta_{cw}$: wild-type cells are encountered with frequency f_W , and the cancer cell wins the Moran competition with probability β_{cw} .
- $\sum_j f_j \beta_{cc}$: other cancer cells are encountered with total frequency $\sum_j f_j$, with a symmetric 50% win rate $\beta_{cc} = 1/2$.

Elementary transitions

We now enumerate every possible microscopic event and its rate. The key insight is that during a single division event of a class- k cancer cell, two independent fates are resolved: (i) the **mother fate** (what happens to the original cell on its site after mutation), and (ii) the **daughter fate** (what happens to the copy placed at the target site after mutation). Both fates are conditioned on the division being successfully completed.

The complete transition rate matrix is given in Table 2. We explain each row below.

Table 2: Complete Microscopic Transition Rate Matrix (Symmetrical Hazards)

Active Vector	Target & Event Type	ΔC_k	ΔD	Transition Rate (λ)
Cancer Class k	Mother Fate: Dies on Site	-1	+1	$C_k r_{\max} \phi_k \delta_k$
Cancer Class k	Mother Fate: Promotes on Site ($k \rightarrow k+j$)	$-1_k, +1_{k+j}$	0	$C_k r_{\max} \phi_k P_{k \rightarrow k+j}$
Cancer Class k	Daughter Fate: Clones on D	+1	-1	$C_k r_{\max} p_D \alpha_k$
Cancer Class k	Daughter Fate: Promotes on D ($k \rightarrow k+j$)	$+1_{k+j}$	-1	$C_k r_{\max} p_D P_{k \rightarrow k+j}$
Cancer Class k	Daughter Fate: Dies on D	0	0	$C_k r_{\max} p_D \delta_k$
Cancer Class k	Daughter Fate: Clones on W	+1	0	$C_k r_{\max} f_W \beta_{cw} \alpha_k$
Cancer Class k	Daughter Fate: Dies on W	0	+1	$C_k r_{\max} f_W \beta_{cw} \delta_k$
Wild-Type (W)	WT Divides, Replaces Dead Void	0	-1	$W r_0 p_D$
Wild-Type (W)	WT Divides, Clears Cancer k	-1	0	$W r_0 f_k \beta_{wc}$

1.2 Mean-Field Ordinary Differential Equations

By taking the thermodynamic limit ($N \rightarrow \infty$), the stochastic master equation converges to a deterministic system of ODEs. We derive the ODE for each compartment by summing all transition rates from Table 2 that affect that compartment, converting from absolute counts to densities ($f_k = C_k/N$).

Interior cancer classes ($1 < k < N_I$)

For an interior cancer class k ($1 < k < N_I$), the rate of change of the cell fraction f_k is determined by five primary biological processes: (i) net clonal growth and survival ($r_{\max} f_k \phi_k (2\alpha_k - 1)$), which balances faithful daughter cloning against mother cell loss via death or promotion; (ii) self-displacement loss within the same class ($r_{\max} f_k^2/2$); (iii) cross-class loss due to displacement by other cancer classes ($f_k \sum_{j \neq k} r_{\max} f_j \beta_{cc}$); (iv) wild-type clearance loss ($r_0 f_W f_k \beta_{wc}$); and (v) promotion influx from all upstream classes $i < k$ ($\sum_{i=1}^{k-1} 2 r_{\max} f_i \phi_i P_{i \rightarrow k}$). Combining these contributions yields:

$$\frac{df_k}{dt} = r_{\max} f_k \phi_k (2\alpha_k - 1) - r_{\max} \frac{f_k^2}{2} - f_k \sum_{j \neq k} r_{\max} f_j \beta_{cc} - r_0 f_W f_k \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i \phi_i P_{i \rightarrow k} \quad (12)$$

Boundary class $k = 1$

The founding cancer class $k = 1$ receives no upstream mutational influx (there is no class $k = 0$). The equation is identical to the interior case but without Term 6:

$$\frac{df_1}{dt} = r_{\max} f_1 \phi_1 (2\alpha_1 - 1) - r_{\max} \frac{f_1^2}{2} - f_1 \sum_{j \neq 1} r_{\max} f_j \beta_{cc} - r_0 f_W f_1 \beta_{wc} \quad (13)$$

Terminal class $k = N_I$

For the terminal mutational class, all instability genes are already mutated, so further promotion is impossible: $P_{N_I \rightarrow N_I+j} = 0$. The mother-loss term simplifies to

$-r_{\max}f_{N_I}\phi_{N_I}(1 - \alpha_{N_I})$ where $1 - \alpha_{N_I} = \delta_{N_I}$. The promotion influx from all classes $i < N_I$ is present:

$$\frac{df_{N_I}}{dt} = r_{\max}f_{N_I}\phi_{N_I}(2\alpha_{N_I}-1) - r_{\max}\frac{f_{N_I}^2}{2} - f_{N_I} \sum_{j \neq N_I} r_{\max}f_j\beta_{cc} - r_0f_Wf_{N_I}\beta_{wc} + \sum_{i=1}^{N_I-1} 2r_{\max}f_i\phi_iP_{i \rightarrow N_I} \quad (14)$$

Wild-type compartment f_W

The WT population grows when WT cells divide and successfully place daughters on dead slots or displace cancer cells, and shrinks when cancer daughters displace WT cells:

- **Gain:** WT divides onto dead slots (rate $r_0f_Wp_D$) or displaces cancer class j (rate $r_0f_Wf_j\beta_{wc}$ per class).
- **Loss:** Cancer class k daughters displace WT cells (rate $r_{\max}f_kf_W\beta_{cw}$ per class).

$$\frac{df_W}{dt} = \underbrace{r_0f_W \left(p_D + \sum_{j=1}^{N_I} f_j\beta_{wc} \right)}_{\text{WT expansion (dead slots + cancer clearance)}} - \underbrace{f_W \sum_{k=1}^{N_I} r_{\max}f_k\beta_{cw}}_{\text{displacement by cancer daughters}} \quad (15)$$

Dead compartment p_D

Dead cells accumulate from lethal mutations (both mother dying in place and daughter dying after placement) and are consumed when any living cell fills a dead slot. Since DNA replication only happens during successful division, both mother and daughter can only die if division succeeds.

- **Gain from mother death:** $r_{\max}f_k\phi_k\delta_k$ per class (row 1).
- **Gain from daughter death:** $r_{\max}f_k\phi_k\delta_k$ per class (combining rows 5, 7, and cancer-cancer targets).
- **Total death-related gain:** $2r_{\max}f_k\phi_k\delta_k$ per class.
- **Loss from filling:** Dead slots are consumed by cancer daughters ($r_{\max}f_kp_D$, regardless of fate) and WT daughters ($r_0f_Wp_D$).

$$\frac{dp_D}{dt} = \underbrace{\sum_{k=1}^{N_I} r_{\max}f_k [2\phi_k\delta_k - p_D]}_{\text{cancer-driven death accumulation minus filling}} - \underbrace{r_0f_Wp_D}_{\text{WT filling dead slots}} \quad (16)$$

1.3 Stationary State Distributions and Stability Analysis

Linear stability of the healthy fixed point

We examine the linear stability of the healthy fixed point ($f_W^* = 1$, $p_D^* = 0$, $f_k^* = 0 \forall k$) against an infinitesimal perturbation $f_1 = \epsilon \ll 1$, with all other cancer classes still at zero.

Starting from the $k = 1$ boundary ODE and substituting the healthy fixed-point values, each term simplifies as follows:

- **Mother loss & daughter clone (combined):** $r_{\max} \epsilon \phi_1 (2\alpha_1 - 1)$. At the fixed point, $\phi_1 \rightarrow f_W \beta_{cw} = 1 \cdot \beta_{cw}$. This gives $r_{\max} \beta_{cw} (2\alpha_1 - 1) \epsilon$.
- **Self-displacement:** $-r_{\max} \epsilon^2/2 \approx 0$ — vanishes at $\mathcal{O}(\epsilon^2)$.
- **Cross-class loss:** $-\epsilon \sum_{j \neq 1} r_{\max} f_j \beta_{cc} = 0$.
- **WT clearance:** $-r_0 (1) \epsilon \beta_{wc} = -r_0 \beta_{wc} \epsilon$.
- **Promotion influx:** absent for $k = 1$.

Collecting all $\mathcal{O}(\epsilon)$ terms:

$$\frac{df_1}{dt} \approx [r_{\max} \beta_{cw} (2\alpha_1 - 1) - r_0 \beta_{wc}] \epsilon \quad (17)$$

Substituting $\beta_{cw} = r_{\max}/(r_{\max} + r_0)$ and $\beta_{wc} = r_0/(r_{\max} + r_0)$:

$$\frac{df_1}{dt} \approx \frac{r_{\max}}{r_{\max} + r_0} [r_{\max}(2\alpha_1 - 1) - r_0] f_1 \quad (18)$$

Derivation of the growth foothold inequality

The tumor can invade when the bracketed growth rate is positive:

$$r_{\max}(2\alpha_1 - 1) - r_0 > 0 \quad (19)$$

This yields the Unified Growth Foothold Inequality:

$$r_{\max}(2\alpha_1 - 1) > r_0 \quad (20)$$

The left-hand side balances the cancer cell's maximum birth rate times its net clonal expansion factor $(2\alpha_1 - 1)$, which accounts for the survival and fidelity of both the mother and daughter during a division event. The right-hand side r_0 represents the WT homeostatic division pressure. When mutation rates are too high, the cancer's net growth rate drops below the wild-type's division rate, preventing the tumor from taking foothold.

Stationary state

When the growth inequality is satisfied, the system reaches a tumor-dominated stationary state where $df_k/dt = 0$ for all k . Setting the interior ODE to zero and rearranging:

$$0 = r_{\max} f_k^* \phi_k^* (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \quad (21)$$

We separate the ϕ_k^* sum into its self-interaction part ($f_k^* \beta_{cc}$) and the rest ($\phi_k^{(\neq k)*}$):

$$\phi_k^* = \phi_k^{(\neq k)*} + f_k^* \beta_{cc} \quad (22)$$

Substituting and grouping terms by powers of f_k^* :

$$\begin{aligned} 0 &= r_{\max} f_k^* \left[\phi_k^{(\neq k)*} + f_k^* \beta_{cc} \right] (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \\ 0 &= r_{\max} f_k^* \phi_k^{(\neq k)*} (2\alpha_k - 1) + r_{\max} (f_k^*)^2 \beta_{cc} (2\alpha_k - 1) - r_{\max} \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \end{aligned} \quad (23)$$

Since $\beta_{cc} = 1/2$, the quadratic terms combine as:

$$-r_{\max} (1 - \alpha_k) (f_k^*)^2 \quad (24)$$

Collecting terms, this takes the standard quadratic form:

$$A_k (f_k^*)^2 - B_k f_k^* - Q_{k-1}^* = 0 \quad (25)$$

where the coefficients are:

1. Quadratic (self-interaction) coefficient (A_k):

$$A_k = r_{\max} (1 - \alpha_k) \quad (26)$$

This arises from the combination of self-displacement ($\beta_{cc} = 1/2$) and the mutation outcomes.

2. Linear (net growth) coefficient (B_k):

$$B_k = r_{\max} \phi_k^{(\neq k)*} (2\alpha_k - 1) - \sum_{j \neq k} r_{\max} f_j^* \beta_{cc} - r_0 f_W^* \beta_{wc} \quad (27)$$

This coefficient represents the net per-capita growth rate of class k at the stationary state, excluding self-interaction and upstream promotion.

3. Upstream promotion source (Q_{k-1}^*):

$$Q_{k-1}^* = \sum_{i=1}^{k-1} 2 r_{\max} f_i^* \phi_i^* P_{i \rightarrow k} \quad (28)$$

This is the rate at which cells flow into class k from all lower classes $i < k$ (dual-sourced: mother promoting + daughter promoting). For $k = 1$, $Q_0^* = 0$.

Solution via the quadratic formula

The quadratic $A_k x^2 - B_k x - Q_{k-1}^* = 0$ (with $x = f_k^*$) has the physical (positive) root:

$$f_k^* = \frac{B_k + \sqrt{B_k^2 + 4 A_k Q_{k-1}^*}}{2 A_k} \quad (29)$$

The positive sign before the square root is chosen because $f_k^* \geq 0$ is required. The system is solved sequentially: f_1^* is determined first (with $Q_0^* = 0$), then f_2^* using f_1^* , and so on up to $f_{N_I}^*$.

2 Simulation Parameters

Here are presented the key parameter values and configurations used in the simulation framework, both in the neighborhood-restricted solid tumor model and the global, well-mixed liquid tumor model.

2.1 Genomic Architecture and Baseline Rates

Table 3 outlines the shared genomic architecture and rate parameters that are identical across both the solid and liquid tumor models to ensure direct comparability.

Table 3: Shared Genomic Architecture and Baseline Simulation Rates

Parameter	Symbol	Default Value	Biological / Model Description
<i>Genomic Architecture</i>			
Ploidy Copy Number	N_{CHR}	2 ^a	Initial number of chromosome copies per cell
Instability Genes	N_I	10	Loci driving mutation rate increments
Oncogenes	N_O	10	Loci driving division rate increments
Suppressor Genes	N_S	10	Loci driving division rate increments
Missegregation Genes	N_M	5	Loci driving chromosome missegregation
Housekeeping Genes	N_{HK}	10	Essential loci (homozygous mutation is lethal)
Total Loci per Chromosome	N_{genes}	45	Sum of all gene classes ($N_I + N_O + N_S + N_M + N_{HK}$)
Initial Cancer Seed State	–	–	1 mutated I gene (bit 0), 1 mutated O gene (bit 10) ^b
<i>Baseline Rates</i>			
Baseline Division Rate	r_0	0.15	Birth rate of healthy wild-type cells ($r_{\text{WT}} = r_0$)
Driver Rate Increment	dr	0.008	Birth rate increase per mutated O or S allele
Maximum Division Rate	r_{max}	0.30	Upper cap on division rate ($2 \times r_0$)
Baseline Mutation Rate	μ_0	0.0	Background mutation rate per locus per division

^a Set to $N_{\text{CHR}} = 1$ (haploid) specifically during the stability sweeps.

^b Represented as chromosome bitmask where all other 43 loci start unmutated.

2.2 Simulation Configurations and Sweep Parameters

Table 4 details the distinct parameter sets and tissue environments used for the four primary studies in both the Solid and Liquid cases: Ensemble Simulations, Stability Sweeps, Parameter Phase Diagrams, and Therapeutic Interventions.

Table 4: Systematic Overview of Simulation Configuration Parameters

Configuration Variable	Ensemble Simulations	Stability Sweeps	Parameter Phase Diagrams	Therapeutic Interventions
Solid Tumor Model (Local Moran-like competition in 2D Moore neighborhood)				
Lattice Side (L)	200	200	80	200
Total Cells ($N = L^2$)	40,000	40,000	6,400	40,000
Max Simulation Steps (n_{steps})	2500	500	1000	Up to 3000 ^a
Ploidy (N_{CHR})	2	1 (Haploid)	2	2
Seeding Strategy	Circular seed ($R = 0.005L$)	Circular seed ($R \approx 0.252L$)	Circular seed ($R = 0.05L$)	Circular seed ($R = 0.05L$)
Initial Tumor Fraction ($f_C(0)$)	$\sim 0.01\%$ (~ 5 cells)	20.0% ($\sim 8,000$ cells)	$\sim 0.8\%$ (~ 49 cells)	$\sim 0.8\%$ (~ 314 cells)
Mutation Rate Increment ($d\mu$)	0.012	Swept $[10^{-6}, 0.02]^b$	Swept $[0.0001, 0.03]$	0.012 (0.024 in D)
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	0.0
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	Whole
Replicate Count	500 per ploidy	1 per r_{max} step	50 per grid point (20×20)	1 per intervention
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.76$ or $f_{\text{WT}} > 0.84$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$
Liquid Tumor Model (Global Moran-like competition with random placement)				
Lattice Side (L)	200	200	80	— ^c
Total Cells ($N = L^2$)	40,000	40,000	6,400	—
Max Simulation Steps (n_{steps})	2500	3000	3000	—
Ploidy (N_{CHR})	2	1 (Haploid)	2	—
Seeding Strategy	Scattered random sites	Scattered random sites	Scattered random sites	—
Initial Seed Count (n_{seed})	10 cells	10 cells	50 cells	—
Mutation Rate Increment ($d\mu$)	0.023 (D) / 0.045 (A, P)	Swept $[10^{-6}, 0.04]^b$	Swept $[0.0001, 0.03]$	—
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	—
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	—
Replicate Count	500 per ploidy	1 per r_{max} step	50 per grid point (20×20)	—
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_C = 1.0$ or $f_C = 0.0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	—

^a Pre-intervention phase runs up to 2000 steps until the tumor density $f_C \geq 0.30$ trigger is met; the post-intervention phase runs for exactly 1000 steps.

^b Refined dynamically using a bisection algorithm with 14 iterations for each division rate step.

^c Therapeutic interventions are implemented and analyzed exclusively for the solid tumor model.

Note on Ensemble Labels: **D** = Diploid ($N_{\text{CHR}} = 2, dm = 0.0$), **A** = Aneuploid ($N_{\text{CHR}} = 2, dm = 0.01$, chunk-based missegregation), **P** = Polyploid ($N_{\text{CHR}} = 2, dm = 0.01$, whole-chromosome missegregation).

2.3 Therapeutic Intervention Mechanisms (Solid Tumor)

Table 5 describes the targeting rules, execution parameters, and physiological analogues of the four distinct therapeutic interventions simulated in the solid tumor model.

Table 5: Mechanistic Parameters of simulated Therapeutic Interventions

Type	Biological Concept	Model Parameter	Clearance / Mutation Update Rule
A	Fitness-selective targeted therapy	r_i (cell division rate)	Active cancer cells are cleared (reset to WT) with cell-specific probability $P(\text{clear}) = 1 - r_0/r_i$.
B	Driver-selective early clone therapy	$i_{\text{lim}} \leq 1$ ($\mu \leq d\mu$)	Cells belonging to mutation class 0 or 1 are cleared with probability $p_d = 1.0$.
C	Instability-selective late clone therapy	$i_{\text{lim}} \geq 3$ ($\mu \geq 3d\mu$)	Cells belonging to mutation class ≥ 3 are cleared with probability $p_d = 1.0$.
D	Mutagenic load therapy	new $d\mu = 0.024$	The mutation rate increment parameter $d\mu$ is doubled from 0.012 to 0.024, immediately doubling the mutation rates of all living cells.

Common Parameters: Interventions are triggered exactly when the active cancer density $f_C \geq 0.30$ (30% of the tissue area). Displaced or cleared cancer cells are reset to the healthy wild-type state ($r_{\text{cell}} = r_0$, $\mu_{\text{cell}} = \mu_0$, $N_{\text{CHR}} = 2$, chromosomes unmutated).

3 Bifurcation and Critical Transition Analysis

Table 6 details the specific simulation parameters, sweep ranges, tissue capacities, and observed critical thresholds for the bifurcation, extinction time, and critical slowing down analyses presented in Figure 1 of the main manuscript.

Table 6: Simulation Parameters and Critical Values for Bifurcation Analyses (Figure 1)

Panel & Model	Swept Parameter	Sweep Range	Critical Threshold	Dynamical Feature / Regime
(a) Solid $\Delta\mu$	Mutation increment $\Delta\mu$	$[13.0, 21.0] \times 10^{-3}$	$\Delta\mu_c \approx 16.8 \times 10^{-3}$	Continuous transition; critical slowing down peak
(b) Liquid $\Delta\mu$	Mutation increment $\Delta\mu$	$[19.0, 27.0] \times 10^{-3}$	$\Delta\mu_c \approx 22.8 \times 10^{-3}$	Bistability window $[22.7, 23.7] \times 10^{-3}$; slowing down
(c) Solid Δr	Division increment Δr	$[1.0, 7.0] \times 10^{-3}$	$\Delta r_c \approx 3.9 \times 10^{-3}$	Inverse transition (extinction to persistence)
(d) Liquid Δr	Division increment Δr	$[1.0, 4.0] \times 10^{-3}$	$\Delta r_c \approx 2.5 \times 10^{-3}$	Lower persistence threshold under global mixing

Shared Parameters for Bifurcation Runs: Tissue capacity $N = 6,400$ ($L = 80 \times 80$ grid for Solid; global Moran pool for Liquid); initial tumor seed $n_{\text{seed}} = 10$ cancer cells carrying I_1 and O_1 mutations; ploidy $N_{\text{CHR}} = 2$ (diploid); maximum simulation duration $n_{\text{steps}} = 100,000$ steps; extinction threshold $N_{\text{cells}} \leq 40$ ($f_C \leq 0.00625$); replicate count = 10 stochastic realizations per parameter step.

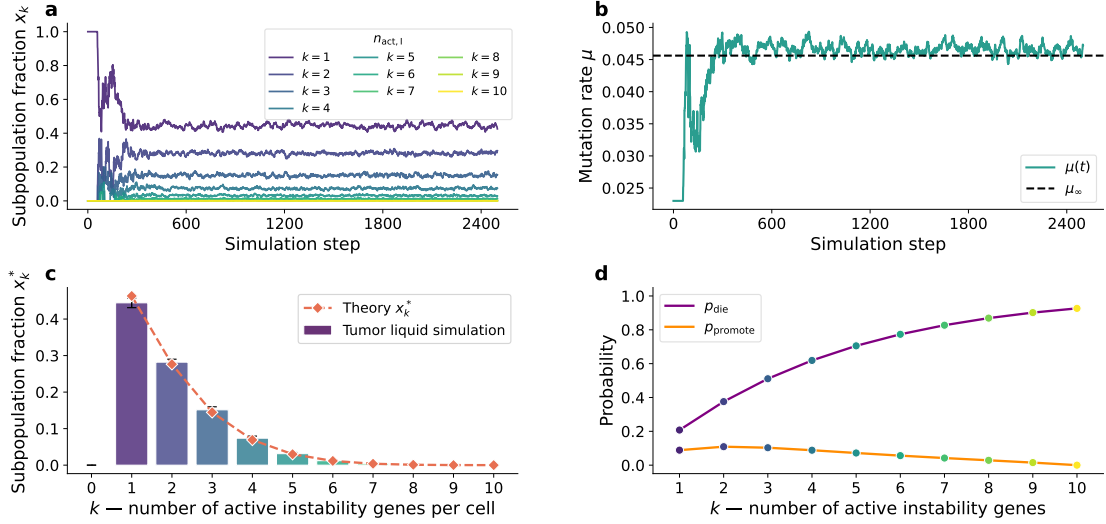


Figure 1: Evolutionary dynamics and steady-state distribution of genetic instability in liquid tumor simulations. (a) Temporal evolution of subpopulation fractions x_k as a function of simulation steps. The population is stratified by k , representing the number of active genetic instability genes per cell ($k \in [1, 10]$). Following initial transient fluctuations, the system relaxes into a stable, non-equilibrium steady state where subpopulations with low genomic instability ($k = 1, 2, 3$) dominate the tumor mass. (b) Dynamic trajectory of the global mutation rate $\mu(t)$ (solid teal line) converging toward the theoretically predicted asymptotic limit μ_∞ (dashed black line). The rapid early expansion corresponds to a transient phase of mutational acceleration before stabilizing due to selective pressures. (c) Steady-state subpopulation fractions f_k^* as a function of active instability genes. Simulated empirical distributions derived from the spatial tumor model (colored bars, error bars denote standard deviation across independent realizations) show excellent agreement with the analytical predictions derived from the mean-field framework (Eq. 29) (dashed orange line with diamonds, **Theory** x_k^*). (d) Functional dependence of phenotypic probabilities on the number of active instability genes k .